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Customized Designs and Engineering Led to Breaking Existing Extended Reach Drilling Records in the Unconventional Vaca Muerta Field, Argentina

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Abstract

Exploiting unconventional reservoirs in the Neuquén Basin presents technical challenges during drilling operations that must be addressed in the initial planning stages. One of the main challenges is drilling 6.75in horizontal sections with increasing lengths that fall into the "extended reach drilling" (ERD) category. The goal of this case study is to share the approach taken in the Vaca Muerta field to exceed the existing lateral section length limits and increase reservoir exposure.

The process begins with identifying the existing technical limits and the conditions that contribute to reaching them. Then, we create customized well-trajectory designs with dogleg severities (DLS) that consider the steering limits by lithology intervals, which are tied to the designed bottom hole assembly (BHA) and selected by a detailed simulation of multiple options focusing on overcoming the drilling challenges of different formations in Vaca Muerta. These modifications and improvements must allow us to drill longer laterals while staying within the rig and drilling equipment technical limits, maintaining directional control, optimizing weight transfer and improving drilling times.

Applying the customized trajectory, detailed drilling roadmap and BHA/Bit design modifications have significantly improved the well placement, steering capability and drilling times. Likewise, calibrated torque and drag simulations have allowed for determining the optimal BHA design to ensure effective weight transfer and extend lateral lengths. A hydraulics sensitivity analysis for drilling with rotary steerable systems (RSS) ensures effective hole cleaning and Rates of Penetration[AP1] (ROP), which are inherent to this technology. Additional modifications to the selected RSS and the bit were introduced as well, including, but not limited to, seal-less technology, robustness, bit-to-push-point distance and equipment material. Finally, reducing tortuosity helped to keep torque, BHA vibrations and temperature within technical limits. These modifications, planning criteria, and drilling approach have enabled us to consistently increase the lengths of the lateral sections, even reaching records above 4,500 m (an improvement of >28%), achieved bottom ROPs of over 100 m/hr (an improvement of 50%) and in some cases, we have exceeded averages of 1,000 m per day by up to 70%.

This work will demonstrate how customized trajectories and BHA designs, including technological developments and operational protocols can increase the length of the reservoir sections while staying within

technical equipment limits. These methods can also be adopted in other basins to ensure the consistent fulfillment of ERD well positioning objectives, improve reservoir exposure and optimize execution times in the Neuquén Basin and other regions.

Introduction

Directional drilling has been for several years the solution that has enabled greater hydrocarbon recovery—especially in horizontal wells, where reservoir exposure is significantly enhanced. One of the main challenges in the unconventional Neuquén Basin is the drilling of wells with increasingly extended horizontal sections, which are technically classified as **Extended Reach Drilling**, since the Measured Depth (MD) exceeds more than twice the corresponding True Vertical Depth (TVD).

The aim of this document is to emphasize key factors from the planning stage to ensure success in these operations—ranging from designing trajectories tailored to each field, selecting the appropriate bottom hole assembly components, and analyzing the operational limits of directional tools and drilling rigs. All of this is intended to guarantee the technical and operational viability of these wells while optimizing execution time.

Additionally, the document will evaluate several operational conditions that challenge project success, such as effective hole cleaning, optimal Weight on Bit (WOB) transfer, directional control, well placement and tortuosity, temperature, and others.

General Information

Extended-Reach Drilling (ERD).

Different definitions can be found for ERD-type wells, but the most common refers to those wells whose measured depth (MD) is at least twice (or more) than their true vertical depth (TVD), thus creating a distinction within horizontal wells ([Robello, S. 2016](#)). ERD wells can be classified as follows:

- Reach-to-TVD Ratio <2 : pozos direccionales convencionales (no ERD).
- Reach-to-TVD Ratio >2 to 3: Extended-Reach Drilling.
- Reach-to-TVD Ratio >3 : Ultra- Extended-Reach Drilling (u-ERD).
- Reach-to-TVD Ratio >15 : Extreme Extended-Reach Drilling (x-ERD).

These wells require the use of technologies and engineering solutions that collectively contribute to achieving the demanding targets. For the purposes of this document, the following aspects are highlighted: trajectory optimization, calibrated torque and drag models, properly designed and calibrated hydraulics, the use of motor assisted rotary steerable systems (RSS), and operational protocols based on lessons learned.

Other solutions include proper design of the drilling mud and its rheology, which must be capable of ensuring hole stability—especially in stratigraphic columns such as the Neuquén Basin, where well control is one of the existing challenges in drilling. Correct selection of drilling equipment is crucial, as it must have the capacity to provide the torque, hoisting, and hydraulics required for extended-reach wells.

Common Metrics in the Neuquén Basin

To provide some context, most operators holding concessions in Argentina's Neuquén Basin (July 2025), are [AP2]drilling lateral extensions with a diameter of 6.75 in, geosteering across various levels of the Vaca Muerta formation. These laterals typically range from 3,000 to 3,500 meters in length. Notably, many of these wells qualify as ERD due to their MD/TVD ratio falling between 2.1 and 2.3, with measured depths ranging from 5,800 to 7,000 meters, and true vertical depths between 2,700 and 3,100 meters approximately.

Regarding rate of penetration, two key indicators are commonly used:

- On-bottom ROP: Defined as meters drilled divided by time with the bit at bottom; typically ranges from 50 to 90 meters per hour.
- Daily footage drilled: Total meters drilled per day, including connection and circulation time; averages between 600 to 1,000 meters per day for lateral sections.

Technical Challenges

One of the main limitations of achieving extended reach in productive sections is inclination. The typical well profiles in Vaca Muerta not only involve building angles from vertical with DLS that, in some cases, exceed $8^\circ/30\text{m}$, but also require lateral sections to maintain inclinations ranging between 88° and 91° , where drag and buckling act against reaching the planned TD.

The influence of friction factors in drilling dynamics is also noteworthy. These dimensionless values vary depending on hole geometry, formation lithology, drilling mud, drill string, and well conditions. This coefficient essentially defines the friction between the wellbore and the drill string and tends to increase proportionally with the directional difficulty index of the trajectory (high DLS and high dispersion between them), resulting in high torque, drag, and weight transfer issues.

Fig.1 shows the friction factor values for a population of 91 wells drilled in a specific Vaca Muerta reservoir between 2021 and 2024, with a range recorded from 0.2 to 0.45, with an average value of 0.28. To normalize the presented data these values were acquired after the well's final circulation, before tripping the BHA out to the surface. The dotted line shows a downward trend in the friction factor.

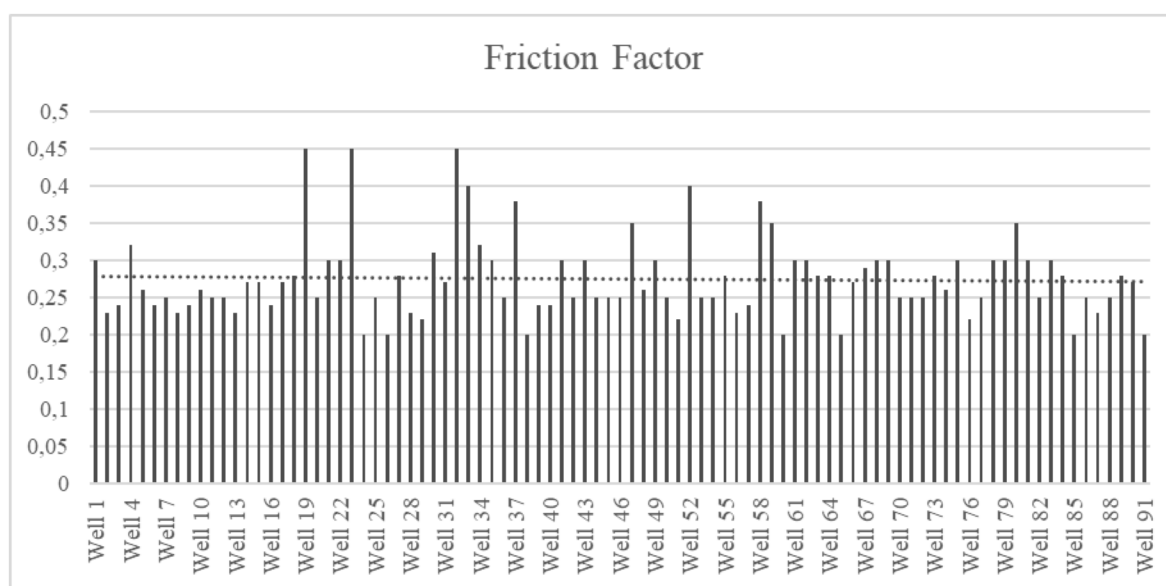


Figure 1—Friction Factor Evolution.

Another limitation that has been increasingly overlooked in recent years is related to the availability of tools and rigs equipped with the technology and capabilities required to drill ERD wells. At this point, it is important to highlight that bottomhole motors were the commonly used tools in the area, both for drilling curve sections and horizontal intervals. However, their directional steering method—which relies on oriented mode (drilling is powered by motor activation through the passage of mud, without the drill string rotating)—hinders the extension a well can reach, due to increased friction that inherently limits the operation's length. In addition, it results in lower ROPs and inefficient hole cleaning. These disadvantages are overcome with the implementation of RSS tools, which rotate 100% of the drill string while steering the trajectory in the desired direction, reducing drag [AP3] friction coefficients and well tortuosity, optimizing weight transfer, and allowing longer lateral sections to be drilled in less time.

On the other hand, non-customized trajectory and BHA designs—without a proper understanding of the formation complexities and their influence on the responses observed in BHAs run in offset wells, and without a timely analysis of lessons learned from prior experiences under similar conditions—may be considered as limiting factors for the success of extended-reach lateral sections.

Trajectory Customization for ERD Wells

Planning is the first and one of the most critical stages in ERD well drilling. At this phase, it is essential to ensure a thorough understanding of the project's requirements and expectations in order to propose a trajectory design that is feasible, practical, and efficient—capable of meeting the required targets, positioning criteria, and the DLS and inclination parameters without compromising either the achievement of the ending point (EP) or subsequent completion and production equipment installation operations.

Trajectory design is influenced by several factors, including those related to the geological complexity and drillability of the formations, as well as those associated with the drill strings such as torque, drag, and buckling. Additionally, well-specific factors must be considered, including wellbore stability, hole cleaning, pipe sticking, casing runs, among others. Finally, collision-avoidance considerations must be addressed, as well as minimizing wellbore tortuosity and optimizing execution times. Therefore, plans are designed to reduce sliding and drilling times in the tangent and lateral sections, while simplifying the number of runs required to drill a well.

In terms of geological complexity, we refer to the UCS (unconfined compressive strength) values of the main formations encountered during the production phase: Quintuco and Vaca Muerta. The mechanical properties of these formations reveal higher average UCS values for the Quintuco Formation, which is generally associated with low directional build-up rates. According to ISRM (International Society for Rock Mechanics and Rock Engineering) terminology, these formations qualify as very high UCS (>2500 bar). [Fig. 2](#) presents UCS information for a reference well, while [Table 1](#) summarizes this data by formation.

Table 1—UCS Average for Quintuco and Vaca Muerta for reference well.

Formation	UCS average, bar	ISRM Standard Terminology
Quintuco	23287	Very high
Vaca Muerta	19426	Very high

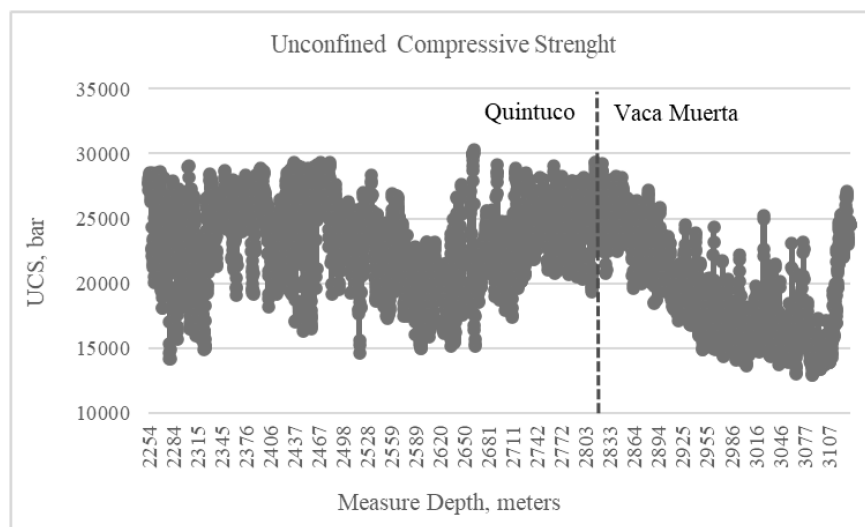


Figure 2—UCS vs MD for reference well.

Typical Well Profiles in Vaca Muerta

Vaca Muerta presents some of the most complex well profiles in the industry:

1. It starts with a surface section that is typically designed vertically or with deviations of up to 8° inclination, cased to isolate the shallow aquifers in the basin.
2. Then, an intermediate section is drilled and depending on the Easting displacement required, may follow an "S" profile — consisting of a build interval up to a certain inclination, followed by a hold interval for a given length, and finally a drop interval to return to vertical. For wells requiring greater horizontal displacements, these sections may follow a "J" profile, with a build up to a determined inclination maintained tangentially through the end of the section (see Fig. 3.). Typically, this section concludes some meters drilled into the Quintuco formation (entry varies depending on field needs) and is cased accordingly for geomechanical reasons, wellbore integrity, and pressure control. In some fields within the basin, this section must be drilled in two stages (Intermediate I and Intermediate II), due to the presence of well-known loss zones in certain shallow formations.
3. Trajectory complexity increases significantly at the final section — the production phase — which generally begins with a vertical segment (or a hold interval if the intermediate profile was type J), followed by a 2D (only build) or 3D (build + turn) curve with DLS typically ranging between $3^\circ/30\text{m}$ and $8.5^\circ/30\text{m}$. This section drills a few meters into the Quintuco formation before entering the transition zone to Vaca Muerta, an area known for its low directional response. Then, it drills into the Vaca Muerta formation, which offers favorable drillability conditions, where landing (LP) is performed at inclinations between 88° and 91° . From there, the lateral section is drilled, which may follow an up dip or down dip trajectory depending on the steering azimuth (North or South) and geosteering decision, until the planned TD.

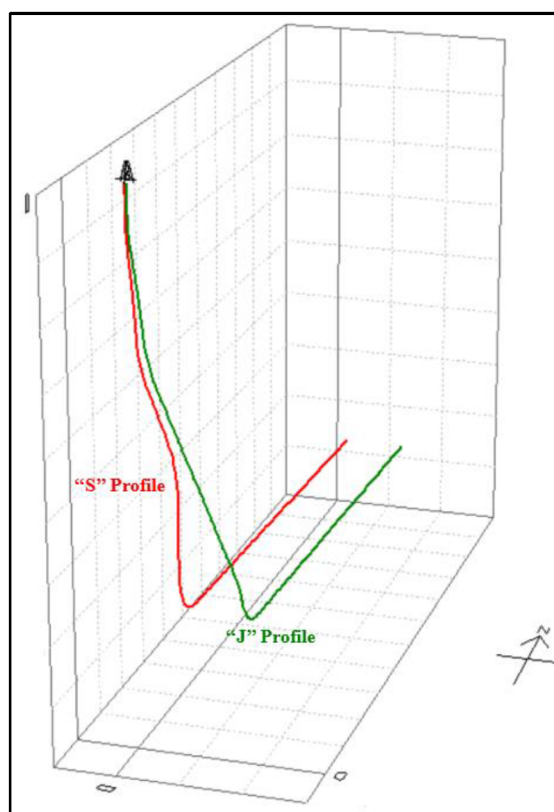


Figure 3—Graphical representation of an "S" type profile vs. a "J" profile.

Motor Yield Evaluation

Motor Yield (MY) is a key tool in trajectory customization, as this metric allows for assessing the effective performance of a BHA during various stages of the curve section and based on the slide meters, relative to the well's inclination angle and depth. These variables are charted to illustrate how the performance of a downhole motor evolves as the curve build progresses, enabling comparison between the actual response achieved for a given interval of slide and the expected response for a 100% slide interval—normalizing the build capability every 30 meters (100 ft).

This graph is used to analyze how the drilled formations and the inclination angle influence the BHA responses, and to apply these insights as a design pattern for trajectories—contributing significantly to improved drilling times and ensuring well positioning in an efficient and effective manner.

Fig. 4. [AP4] displays the response of a directional BHA used to drill a typical curve in Vaca Muerta. The "Y" left axis represents DLS in deg/30m and the "Y" right axis represents the slide interval in meters, while the "X" axis shows TVD in meters (it may also be charted against MD). The red bars are scaled according to the actual DLS obtained at a specific depth; the yellow bars indicate the estimated DLS value assuming a 100% slide interval. The green curve fluctuates according to the slide meters required to achieve each DLS, and the blue curve represents the inclination of the well at each interval. These trends guide the directional design of the production section trajectory in Vaca Muerta.

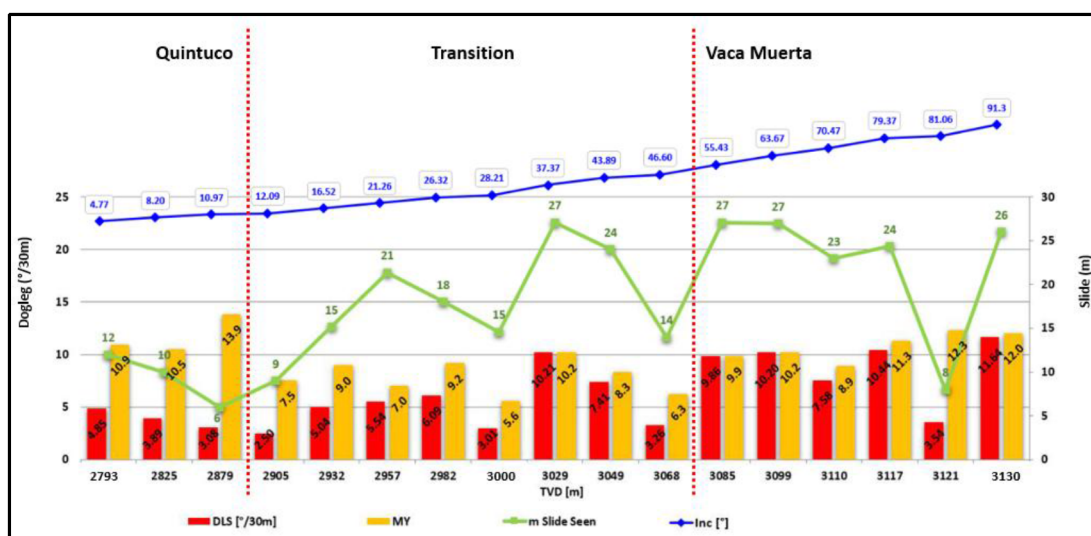


Figure 4—MY chart of a 6.75in curved section, constructed by a 5in downhole motor with a 1.6° bend housing (BH) and a 6.25in sleeve stabilizer.

Customizing Trajectory Designs

The adopted practice consists of the following:

- Surface section validation: determining whether it can remain vertical or if deviation is required, which depends primarily on the available TVD to reach the landing point and the required easting displacement (E/W), with an achievable DLS.
- Intermediate section planning: achieving the E/W displacement is the primary goal, and in some cases, the most challenging aspect of this section. Based on lessons learned, it is recommended to implement a build-up rate (BUR) between 1°/30m and 2.5°/30m, followed by relatively short tangent sections (typical S-shaped profiles with hold intervals ranging between 100m and 200m) since this prevents increasing slide percentages for trajectory control. If a return to vertical is needed, a reduction in execution times have been reached by designing the drop phase based on

the natural rotation drop of the BHA, which varies depending on how pendulum-like the assembly is designed and the drilling parameters established in the road map.

- Production Section Strategy:
 - Curve section geometry should be planned using stepped DLS values instead of a uniform BUR from KOP to the LP. These DLS adjustments are based on MY and formational passes, considering the drillability of each interval. For detailed information on customized trajectory designs in Vaca Muerta curve sections and their impact on drilling performance, refer to SPE paper number 223487-MS (Maita, M.A., Noguera, D., 2024).
 - Within the reservoir, if lateral extension involves azimuth or inclination changes, these adjustments are planned with $DLS \leq 1^{\circ}/30m$ to minimize added tortuosity and reduce the risk of missing the ERD ending point.

As part of the ongoing effort to reduce wellbore tortuosity, it is important to note that in ERD-type wells, the E/W displacement is ideally completed during the intermediate section. However, if it is necessary to complete this displacement within the production section, due to operational factors or limitations in available TVD, the recommendation is to plan directional work as smoothly as possible during the initial meters of the lateral extension, thereby avoiding 3D curves.

Fig. 5. [AP5] illustrates an ERD trajectory design that begins with a vertical surface phase, followed by an intermediate S-type section with a build interval of $2^{\circ}/30m$, then a hold interval and a subsequent drop customized according to lessons learned. This drop includes an initial segment with a DLS of $-1^{\circ}/30m$, followed by a second stage where the BHA naturally decreases inclination while rotating, with a rate of $-0.35^{\circ}/30m$ until returning to vertical.

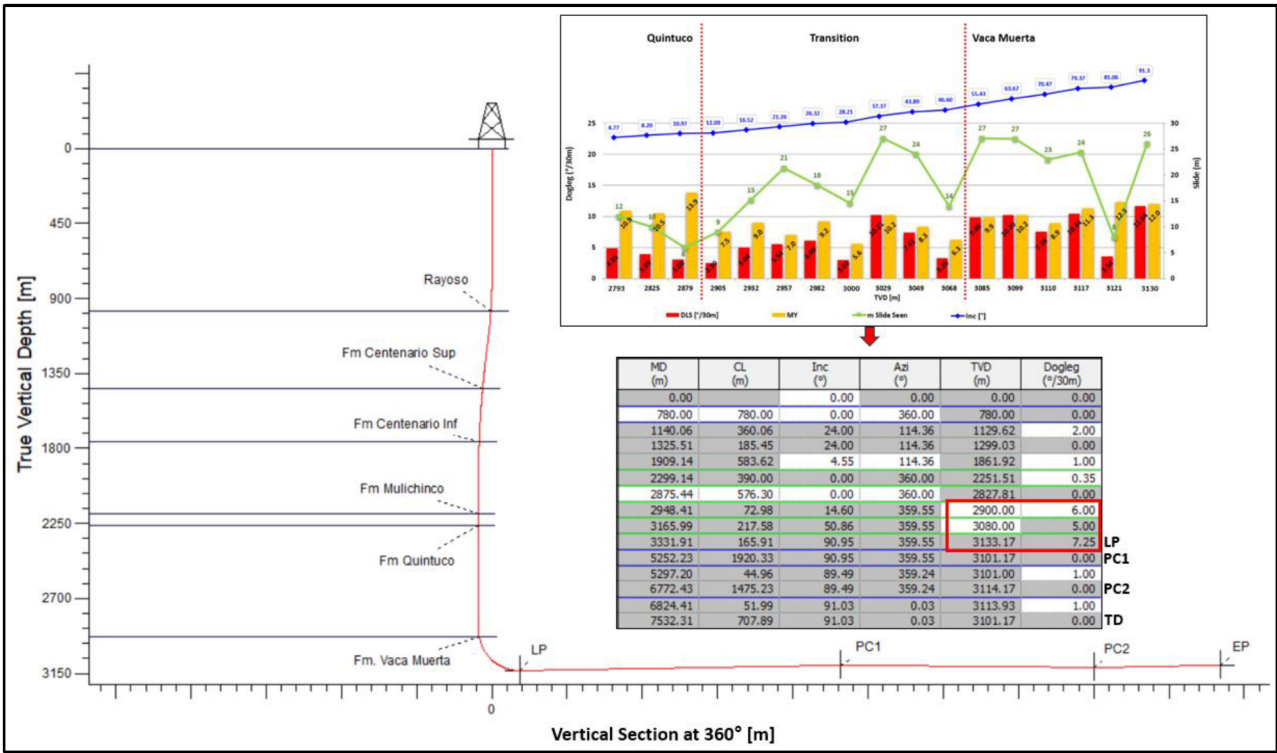


Figure 5—Graphical representation of the trajectory customization based on MY records of offset wells.

The curve section starts with a DLS of $6^{\circ}/30m$ up to the top of the Vaca Muerta formation, followed by a planned BUR reduction to $5^{\circ}/30m$ to pass through the transitional zone and enter Vaca Muerta by a few

meters (the length of this transition tends to be fairly consistent across wells in the same field, especially in cases with extensive experience and data available). Finally, once drillability improves, a DLS of $7.25^\circ/30\text{m}$ is used to land the well. Ideally, an average is used for this final stage based on the best responses observed by the same BHA in previous wells. This customization avoids ending up "behind the curve" and the resulting need to increase sliding operations.

Lastly, the lateral extension is planned to reach control points and the EP in a smooth manner, with DLS values not exceeding $1^\circ/30\text{m}$.

Bottom Hole Assembly Design / Modeling

The BHA design for ERD wells is a complex process that involves several engineering analyses and key simulations to properly select and position their components. The designed BHA must be capable of fulfilling the required objectives for the section to be drilled (build, hold, or drop) in the most effective and efficient way. Directional BHA modeling must be able to estimate the response of various possible configurations under operational parameter variations such as weight on bit (WOB), wellbore inclination, over-gauge or under-gauge hole, and formation tendencies. The BH angle, the stabilization implemented, and its placement have a critical effect on BHA behavior, and it is recommended that models be calibrated to improve estimations in response to potential changes.

Torque and Drag

Torque and drag (T&D) modeling considers factors such as the well trajectory, DLS, drill string configuration, friction factors, among others. These simulations serve various purposes, including:

- Determining whether a well can be drilled and cased.
- Evaluating whether a trajectory needs to be optimized to reduce expected T&D.
- Estimating the available weight on bit [AP6] for both rotary and sliding drilling scenarios.
- Identifying torque and hoisting requirements for drill rig selection.

The main challenge in planning an ERD well is to design a drill string configuration that ensures available WOB to make the necessary directional corrections and achieve the targeted trajectory and the TD. At this stage, it's essential to introduce the concept of "push pipe" which refers to the positioning of the heavy weight drill pipe (HWDP), responsible for transferring weight into the drill string.

Field lessons from the basin have shown that, when drilling horizontal sections exceeding 3,500 meters in length, the most effective push pipe placement for optimal weight transfer—enabling trajectory control without excessive drag—has been between 75° inclination and the LP (at 90°) once the bit reaches TD. This is graphically illustrated in [Fig. 6](#) below.

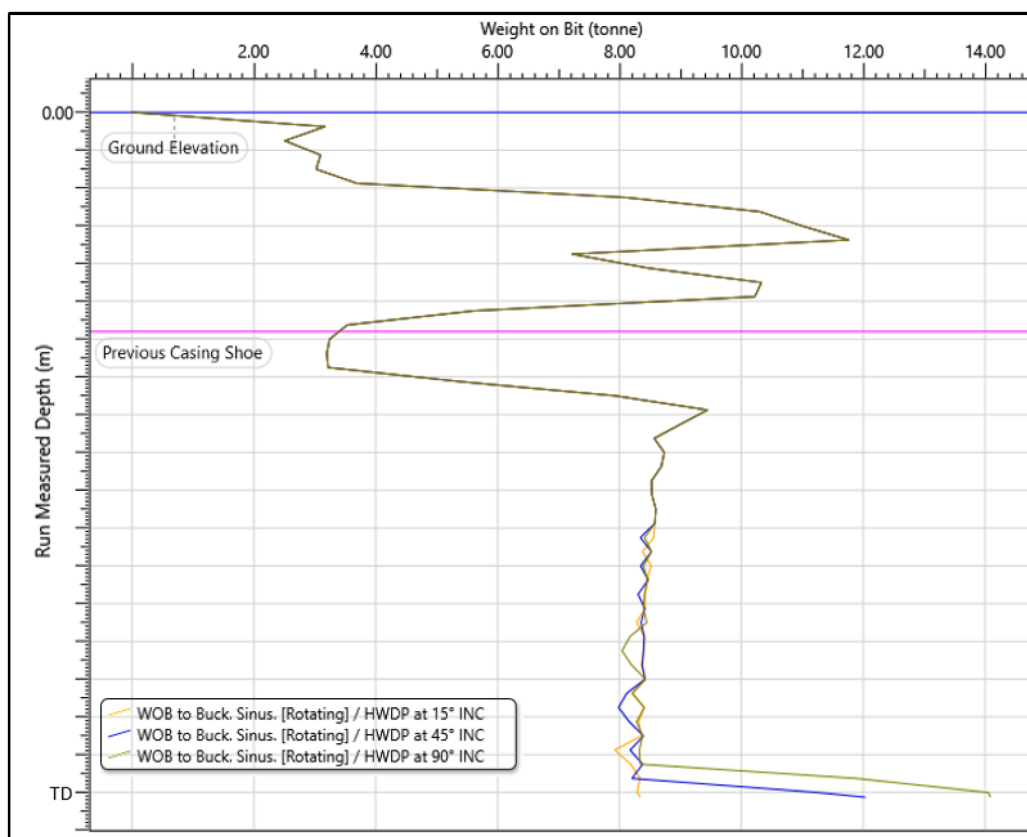


Figure 6—Comparison of the WOB available for drilling in rotation mode, placing the push pipe at 15°, 45° and 90° inclinations.

Another key point worth highlighting is that these extended horizontal sections may require drilling with bottomhole motor BHAs, where WOB transfer during sliding can become critical. Since the drill string must remain stationary while the motor rotates internally, this leads to inefficient weight transfer occurring due to increased drag accumulation—not to mention, the resulting tortuosity with sudden angular changes causes difficulty in reaching the planned TD.

The solution to drilling increasingly challenging lateral lengths has been the use of rotary steerable systems (RSS) across most laterals in the basin. These assemblies have the capability to steer the well in the required direction while rotating 100% of the drill string, generating lower friction factors and consequently less tortuosity, reducing T&D and improving hole cleaning efficiency.

An additional consideration when planning lateral sections exceeding 3,800 meters, is the torque limit of the rig and the drill pipe in use. In such cases, implementing bottomhole motors to assist RSS systems, has proven to be an effective solution to torque limitations, while allowing increased revolutions at the bit (PDC-type), from 120/150 rpm to typically double that (300 rpm), thereby generating an increase in on-bottom ROP.

Hydraulic Modeling

Ensuring effective hole cleaning—despite pressure limitations from drilling equipment—while operators pursue higher ROPs, places hydraulic simulations at the face of ERD operations. Focusing on motor-assisted rotary steerable system (MARSS), which is the best performing technology used for lateral section drilling in Vaca Muerta, a carefully calibrated and sensitive design must be performed. This design must also guarantee the steering capability throughout the entire section to be drilled.

The calibration process relies on data gathered from offset wells that employed similar MARSS BHA configurations under equivalent operational conditions—such as mud density, bit type and total flow area (TFA), flow rates, among others. These inputs allow analysts to evaluate and replicate pressure readings

from the pressure while drilling (PWD) sensors, estimate motor bypass percentages, and ultimately calculate the pressure and optimal force on the RSS pads for directional control.

Armed with this information and aware of the lateral reach challenge ahead, engineers can proceed to design a customized hydraulic plan (see Table 2), where proposed pressure drop configuration should enable flow rates that simultaneously ensure proper hole cleaning and maintain RSS steering capability throughout the section. This reduces overall drilling time while keeping pressure within the operational limits of downhole tools and surface circulation equipment.

Table 2—An example of sensitive hydraulic design for a 6.75-in horizontal ERD section to be drilled using a Motor Assisted RSS (MARSS). This table shows the range of flow rates proposed for drilling a 4,200-meter lateral extension at a given density, with a range of motor bypasses between 8% and 22%. For this hydraulic design, pad forces are expected to range from 1,688 lbf to 1,481 lbf to ensure the desired steer capability along the lateral section within the rig pressure limits.

MD (m)	Surface Flow (gpm)	Flow Split PDM	Flow MWD/RSS (gpm)	MW (kg/m3)	Pad Dif Press (psi)	RSS Press loss (psi)	Calculated Pad Force (lbs)	SPP Off Bottom (psi)
TFA 0.464 in2 (5x11) - RSS MARSS - R23								
3300	260	8.00%	239.20	1750	617.03	489.71	1,688	2900
4000	265	10.00%	238.50	1750	613.42	486.84	1,680	3200
4700	270	12.00%	237.60	1750	608.8	483.18	1,668	3700
5400	270	15.00%	229.50	1750	568	450.79	1,568	4000
6100	275	18.00%	225.50	1750	548.37	435.22	1,519	4400
6800	280	20.00%	224.00	1750	541.1	429.45	1,501	4800
7500	285	22.00%	222.30	1750	532.92	422.95	1,481	5300

Temperature Protocols Considered in Hydraulic Designs

Discussing hydraulic design for an ERD well in Vaca Muerta cannot be done without addressing the temperature protocols that must be implemented toward the end of the lateral sections. This is a formation of shale rich in organic matter, which makes it excellent for hydrocarbon production—but it also has a high heat retention capacity. Due to its geothermal gradient, temperatures typically range between 120°C and 130°C.

On top of that, mechanical friction from drilling and the high revolutions per minute (RPM) at which RSS tools commonly operate contribute additional heat—reaching 150°C or more in certain fields. As such, temperature control protocols are essential to preserve the integrity of directional drilling tools, and these include pumping sweeps of fresh mud, gradually reducing flow rates, and lowering RPM. The latter two actions directly affect the RSS's steering capability and therefore must be considered in the proposed hydraulic program to maintain operational reliability.

Developments / Technical Enhancements in Directional Tools

In preparation for current and future challenges related to ERD wells in the Neuquén Basin, several technological upgrades have been implemented on directional tools, such as:

- Metal-to-metal seals: designed to improve robustness and durability (when compared to elastomeric seals[AP7]). These seals also deliver greater impact force at the pad level in a dual piston design[AP8], thereby enhancing the RSS correction capabilities.
- Robustness: material and proprietary [AP9]connection upgrades in the RSS allow it to withstand the increased stresses typically encountered in these well profiles.
- Bit-to-push-point distance: integrated drill bit design solutions have reduced the distance between RSS pads and the bit, resulting in improved build capability—particularly for trajectories with challenging DLS.

- Equipment material: adaptations to local temperature conditions include new probes and electronics, engineered to resist high temperatures and support extended horizontal drilling sections.

Results Achieved

- These technical improvements in directional tools, design criteria, and operational approach have enabled the successful drilling of full pads with 6.75in ERD wells, reaching MD/TVD ratios up to 2.5, total measured depths ranging from 7,700 to 7,800 meters, and horizontal sections exceeding 4,400 meters.
- Slide percentage reduction and, consequently, improved drilling times. This customized and integrated method for trajectory design and BHA modeling has demonstrated measurable results, with execution times optimized by up to 50% in some cases, slide percentages reduced by 50% or more, while still achieving accurate wellbore placement in target zones and minimizing failure risks.
- Achievement of the longest 6.75-in lateral extension drilled in Vaca Muerta, exceeding 4,500 meters (an improvement of >28%).
- Achieved bottomhole ROPs over 100 m/hr (a 50% improvement), and average daily footage exceeded 1,000 meters per day—an increase of up to 70%.

To explain these changes in the application wells, we refer to Well 1, which was drilled using directional tools without the above-mentioned improvements, and Well 2, which was drilled using these improvements. Although there are no significant differences in the effective ROP between the two wells, the lateral section reached in Well 2 is 17% higher. Up to Q1 of 2025, it will be the longest 6.75in lateral section (4,534 meters) and the longest MD extension (7,828 meters) in the specific Vaca Muerta reservoir studied (Fig. 7 and Fig. 8).

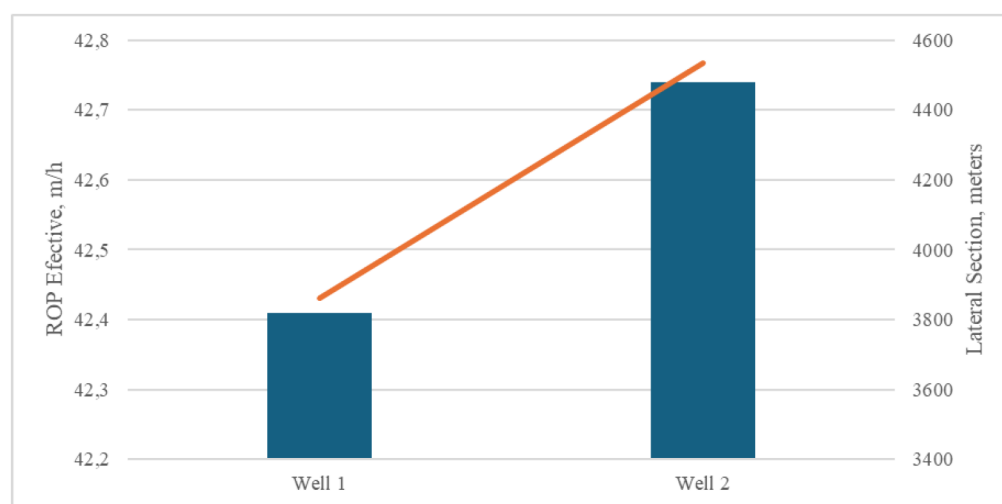


Figure 7—Comparison of ROP Effective and Lateral Section between wells with pre and post Developments / Technical Enhancements in Directional Tools.

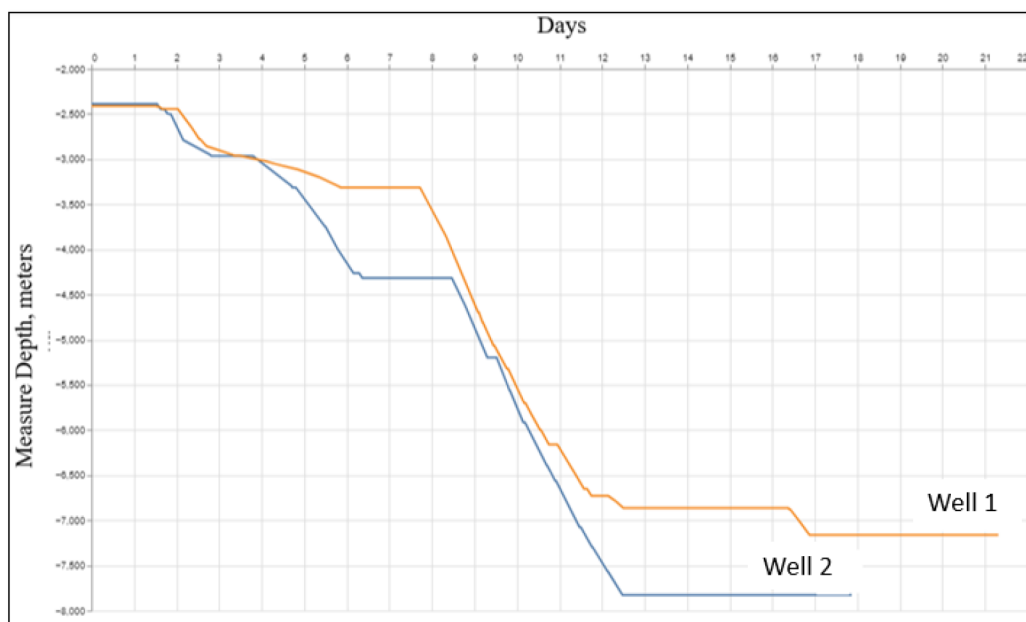


Figure 8—Comparison of Execution Days vs Measure Depth. Well 1 corresponds to an offset well that was drilled without the design methodology explained and without the technical enhancements in directional tools. Well 2, on the other hand, did use the methodology and enhancements.

Focusing only on the production phase, during the execution of these wells shows a favorable difference for Well 2. After technical improvements to the tools, which were implemented in Well 2, a longer lateral section was successfully drilled in one run with RSS. If we compare the rate of days per 100 meters of lateral section, which expresses the time taken to drill 100 meters of lateral per day, we see that this factor decreases by 47% for Well 2 (see Table 3).

Table 3—Comparison of length and days between pre and post wells with Developments / Technical Enhancements in Directional Tools.

Well	LP - EP Days	Lateral Section, meters	Ending Point, meters	LP-EP Days/100 meters of lateral section
Well 1	12,5	3863	7163	0,32
Well 2	7,8	4534	7828	0,17

Conclusions

Successfully achieving the required total depth in ERD wells in the Neuquén Basin demands a comprehensive approach starting in the planning phase. This includes trajectory and BHA designs tailored to each field, technical upgrades to directional tools, and the implementation of operational protocols that—on one hand—preserve the operational limits of directional tools and drilling equipment, and—on the other—ensure consistent well placement, improved reservoir exposure, and optimized execution times.

It is equally important to analyze and address the operational conditions that challenge the success of these types of wells, such as: ensuring effective hole cleaning; maintaining optimal weight transfer to the bit; enabling directional control throughout the entire drilling process with minimal tortuosity; continuously optimizing ROP while managing the elevated temperatures that tools are typically exposed to—thus mitigating the risk of failure.

When applied together, these criteria can significantly contribute to extending the horizontal sections in Vaca Muerta and may also be replicated with similar objectives in other regions around the world.

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